

Kairui Liu

Curriculum vitae | liukairui@pku.edu.cn

Education

2021-2026	Ph.D. in Mathematics Beijing International Center for Mathematical Research, Peking University Advisor: Wenyuan Yang
2017-2021	B.Sc. in Mathematics Peking University

Research interests

Geometric group theory and random walks on discrete groups.

Preprint

Kairui Liu and Wenyuan Yang. *Ancona inequalities along generic geodesic rays*. Preprint, 2025.
[arXiv:2509.11279](https://arxiv.org/abs/2509.11279).

Appendix contribution

Lihuang Ding and Wenyuan Yang. *Growth tightness of quotients by confined subgroups*. Appendix by Lihuang Ding and Kairui Liu. Preprint, 2024. [arXiv:2409.10268](https://arxiv.org/abs/2409.10268).

Invited talks

Martin boundary and geometric boundaries of groups. Geometric Group Theory Seminar, East China Normal University. December 5-7, 2025.

Martin boundary and geometric boundaries of groups. Geometry and Topology Seminar, Peking University. October 29, 2025.

Martin boundary and geometric boundaries of groups. Academic Lecture, Nankai University. October 24, 2025.

Martin boundary and geometric boundaries of groups. The Third International Conference for Young Chinese Probabilists, Xiangtan University. August 8-11, 2025.

Martin boundary and geometric boundaries of groups. Geometry, Group Theory, Topology Seminar, Tsinghua University. May 16, 2025.

Teaching Assistant

Spring 2025	Advanced Mathematics C
Fall 2024	Advanced Mathematics B
Spring 2024	Advanced Mathematics C
Fall 2023	Advanced Mathematics A
Spring 2023	Advanced Mathematics C
Fall 2022	Advanced Mathematics A
Spring 2022	Functional Analysis
Fall 2021	Advanced Mathematics B

Awards and scholarships

2026	Outstanding Graduate (college level)
2024-2025	Outstanding Research Award, Peking University
2024-2025	Lingjun Pilot Scholarship, Graduate Student, Peking University
2015	First Prize (gold medal), 31st Chinese Mathematical Olympiad